

ICT2217

Network Security

Lab 1.3 Notes:

Internal Attacks over LAN and Switch Security Configuration I



Lab Exercise 1.3

1

Double tagging attack

2

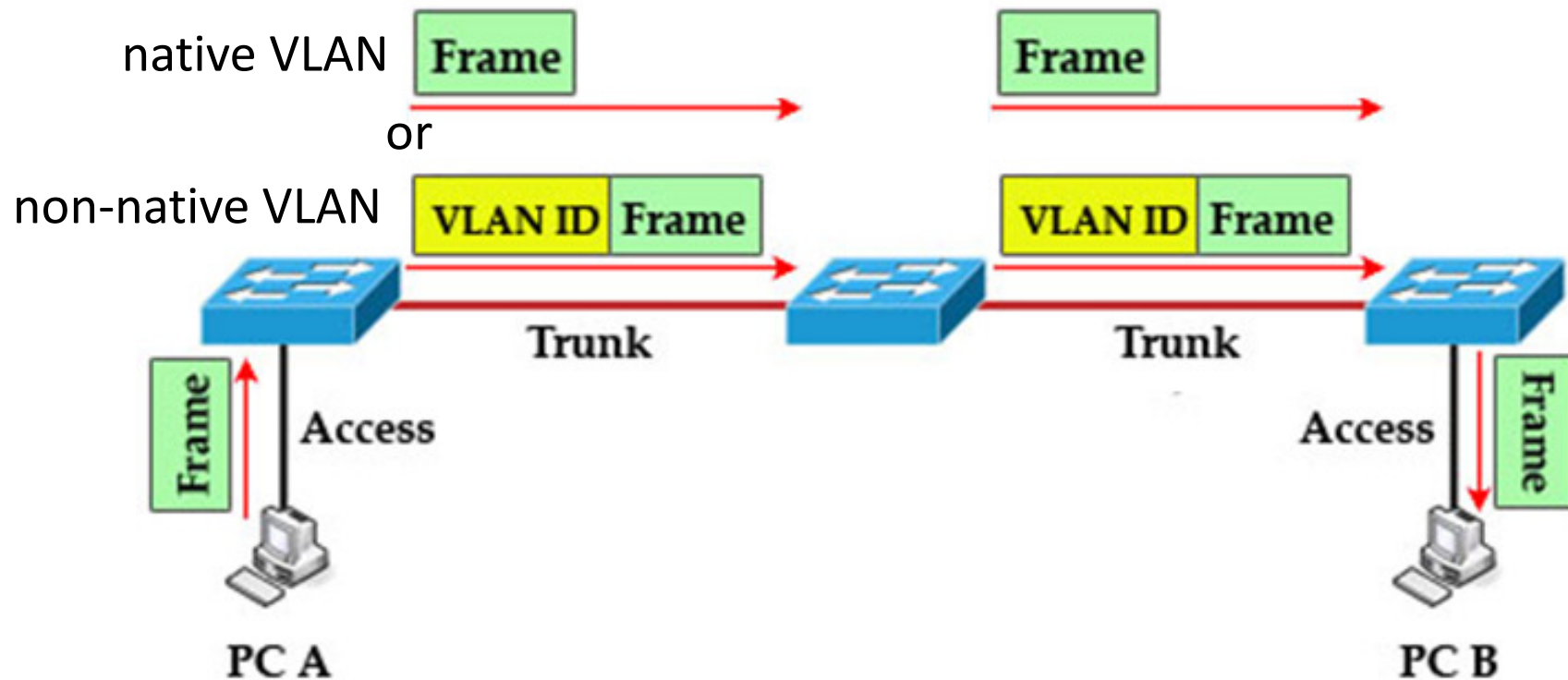
Defenses:

- Configure **dedicated native VLAN ID** for trunk which must not be used for access ports
- Enable tagging for native VLAN if applicable
- Good practice not to use default VLAN 1

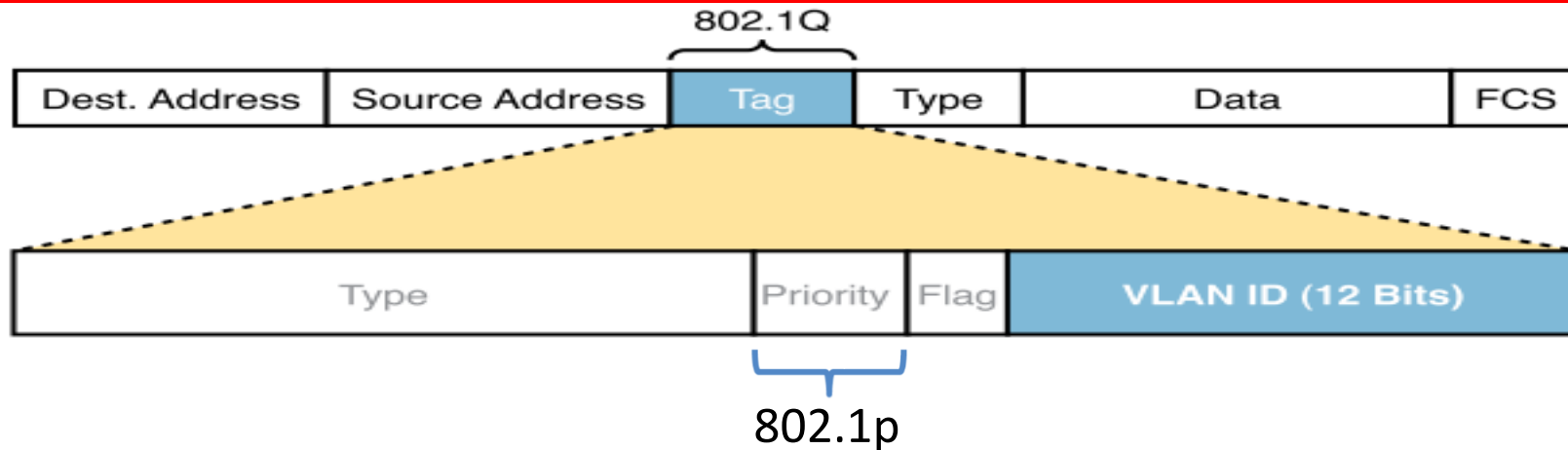
Inside a **trunk**, the different VLAN traffic are distinguished using the **IEEE 802.1Q tagging** standard. (Refer to ICT1013 notes for details.)

Specifically, IEEE 802.1Q supports two types of VLANs inside **trunk**:

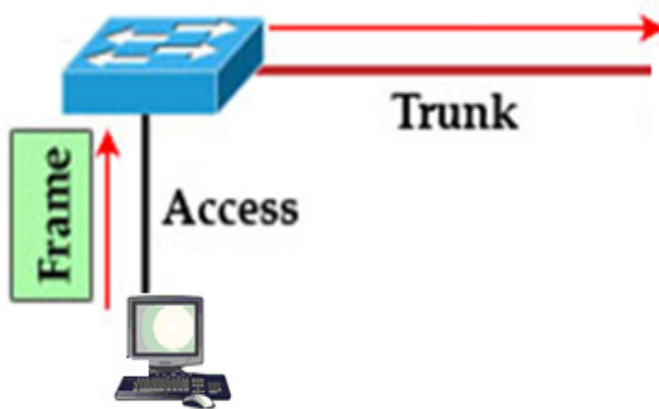
- (i) one **native VLAN** – frames are not tagged
- (ii) all other **VLANs** – frames are tagged with appropriate VLAN ID



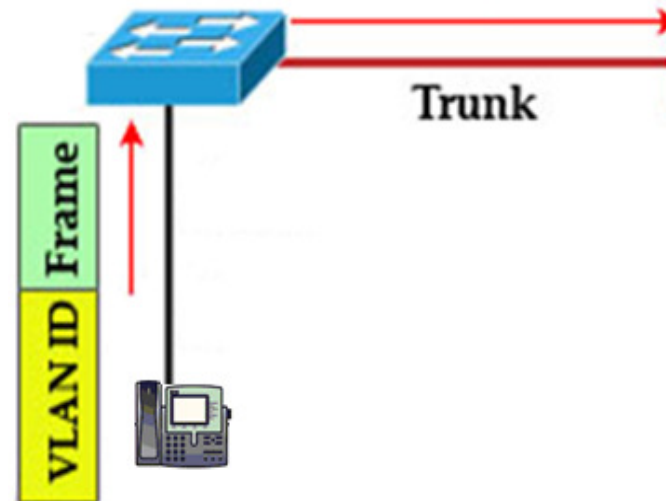
The 802.1Q tag is also being used to support voice over IP (VoIP), specifically the 3-bit priority field is used to specify class of service (CoS) according to IEEE 802.1p standard.



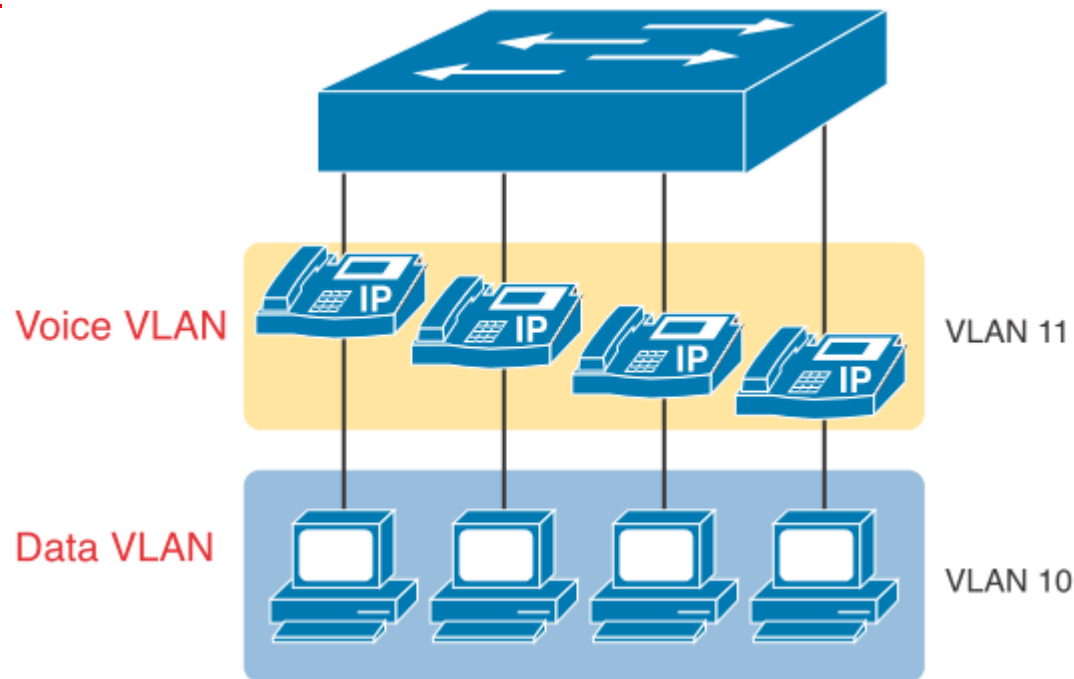
Frames carrying data:



Frames carrying voice:

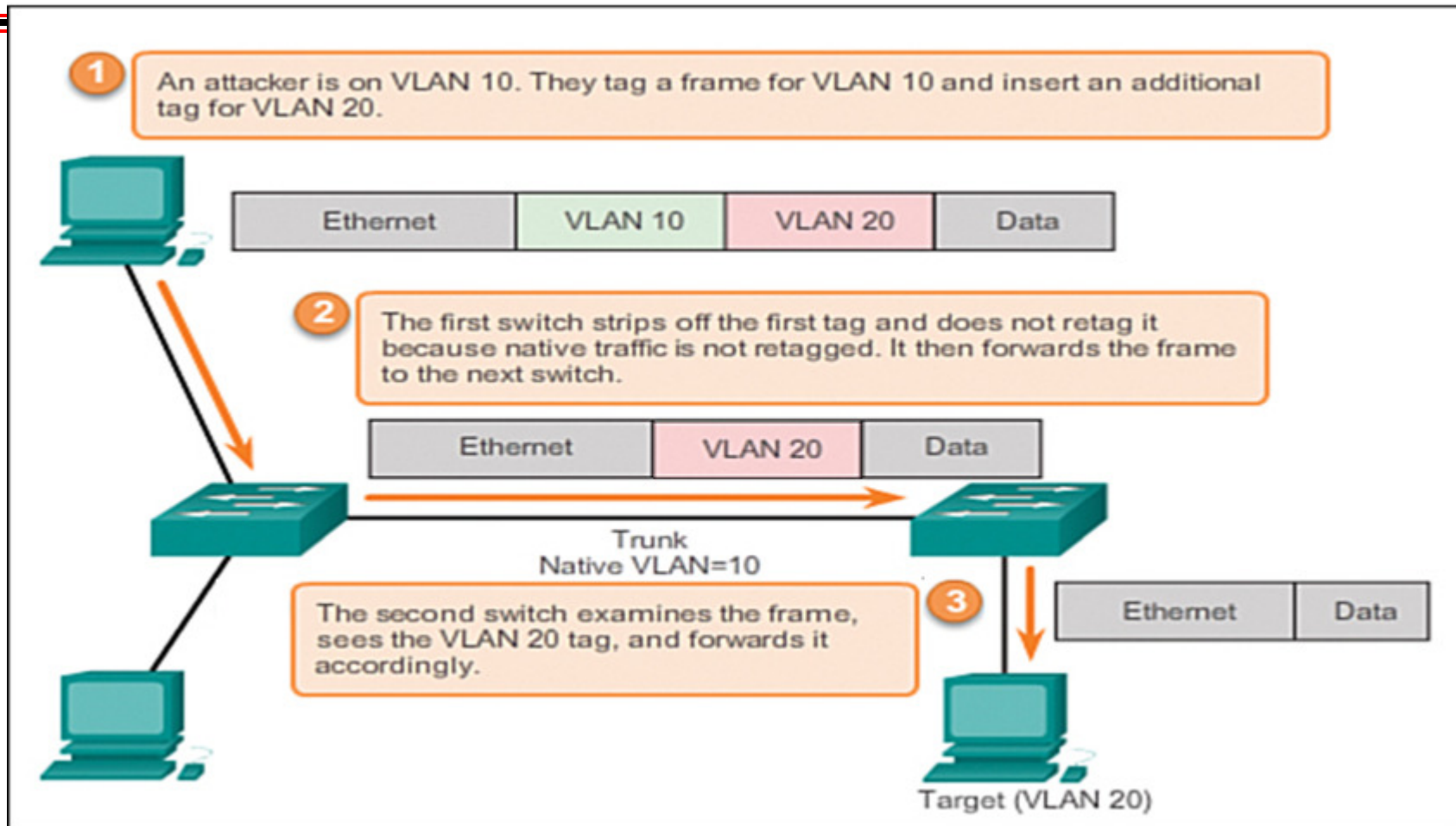


To support **VoIP**, the switch ports will also need to be configured with a data VLAN and an auxiliary **voice VLAN** to receive **tagged frames**.



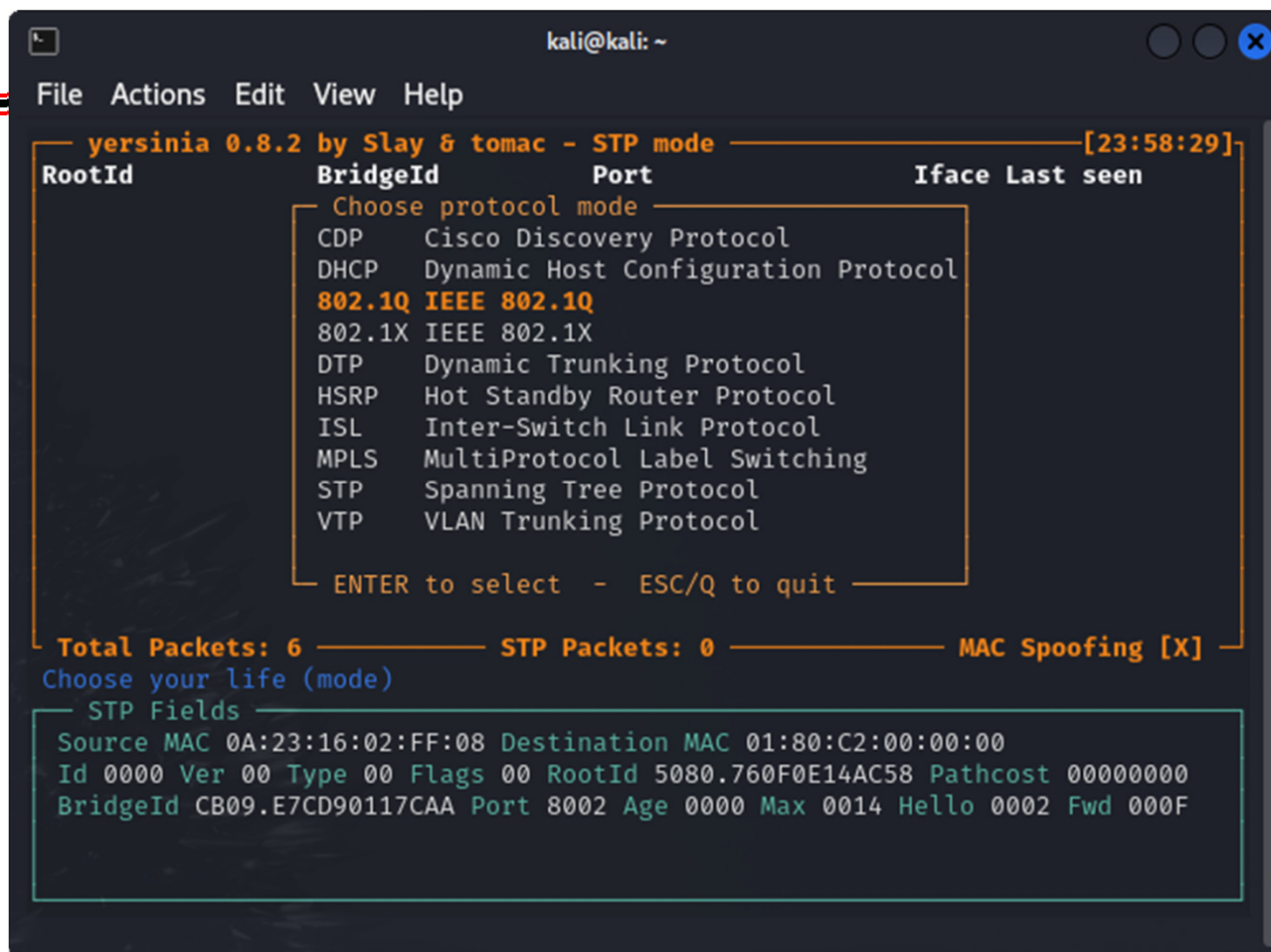
```
SW1(config-vlan)# interface range FastEthernet0/1 - 4
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# switchport voice vlan 11
```

Double tagging attack – another form of VLAN hopping attack where an attacker can send frames to hop to other VLANs without going through a router.



Note: double tagging attack works only if attacker's **access port** has the **same VLAN ID** as the **native VLAN** of the **trunk**, default is VLAN 1.

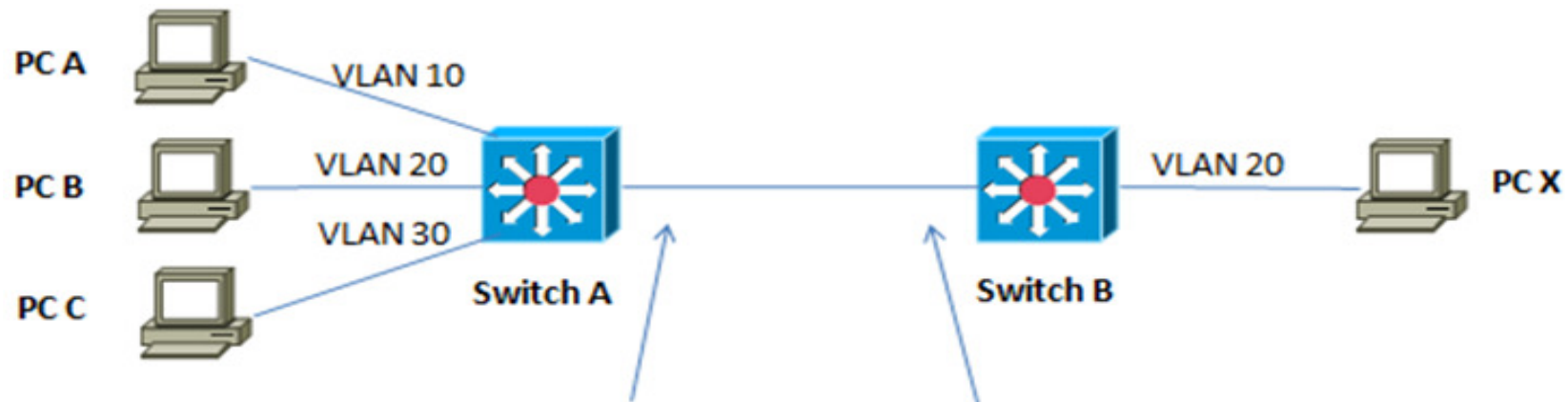
Similarly, Yersinia can be used to perform double tagging attack.



```
kali@kali: ~  
File Actions Edit View Help  
— yersinia 0.8.2 by Slay & tomac - STP mode — [23:58:29]  
RootId      BridgeId      Port      Iface Last seen  
— Choose protocol mode —  
CDP      Cisco Discovery Protocol  
DHCP      Dynamic Host Configuration Protocol  
802.1Q    IEEE 802.1Q  
802.1X    IEEE 802.1X  
DTP      Dynamic Trunking Protocol  
HSRP      Hot Standby Router Protocol  
ISL      Inter-Switch Link Protocol  
MPLS      MultiProtocol Label Switching  
STP      Spanning Tree Protocol  
VTP      VLAN Trunking Protocol  
— ENTER to select - ESC/Q to quit —  
— Total Packets: 6 — STP Packets: 0 — MAC Spoofing [X]  
Choose your life (mode)  
— STP Fields —  
Source MAC 0A:23:16:02:FF:08 Destination MAC 01:80:C2:00:00:00  
Id 0000 Ver 00 Type 00 Flags 00 RootId 5080.760F0E14AC58 Pathcost 00000000  
BridgeId CB09.E7CD90117CAA Port 8002 Age 0000 Max 0014 Hello 0002 Fwd 000F
```

Press key 'g' to display protocol screen, arrow keys to navigate to IEEE 802.1Q and ENTER key to select.

To prevent double tagging attack, configure dedicated native VLAN, e.g. VLAN 500, for all trunk ports which should not be assigned to any access ports.



```
CiscoSwitch(config-if)#switchport trunk native vlan 500
```

Alternatively, new switches like Catalyst 9200L and 9300L in the lab have the feature of forcing all traffic on the trunk to be tagged, including native VLAN.

```
CiscoSwitch(config)#vlan dot1q tag native
```